STAT448-26S1 Assignment 2

15% of your final grade

1 Instructions

Please adhere to the following guidelines for submitting your assignment:

1. Your assignment should be submitted on the Learn platform in one of the following formats: PDF or HTML R-markdown. This format should include written answers, R code, and results, all properly commented.
2. You have the option to complete the assignment in pairs. In such cases, please ensure that both names and student IDs are included on the assignment; BOTH students must submit the assignment.
3. Answer **all** questions. Use the **same** random train/test split for every part of a question (state your $set.seed()$ value clearly at the start of each question).
4. **Reproducibility:** For every random split or cross-validation, use $set.seed(123)$ (or any integer you choose) and document it. Use the exact same $80 : 20$ (or $30 : 12$) split across all relevant parts.
5. **Packages:** Use **glmnet** for all penalized regression (**Ridge** $\alpha = 0$, **LASSO** $\alpha = 1$, **ElasticNet** $0 < \alpha < 1$). Standardise predictors with $scale()$ or $model.matrix$ before fitting any **glmnet** model.
6. **Baseline:** Always report the test-set MSE/RMSE of a plain $lm()$ fit on the same split as a baseline.
7. **Visualisation & tables:** Use **corrplot::corrplot()** or heatmaps, coefficient-path plots ($plot(fit)$), CV-error plots ($plot(cvfit)$), and $knitr :: kable()$ tables for comparisons. A summary table (method, number of predictors kept, test MSE, time, λ/α) is strongly recommended.
8. **Interpretation:** Comment on domain context (e.g., which economic variables matter for housing prices?).
9. Include full reproducibility statement and AI declaration (if used).

10. **Reminder:** Ensure that your submission includes R code, output, and comments for the assigned tasks. These should be presented in a markdown HTML or PDF file. Make sure all relevant code has been run - the markers will not run it for you.
11. **Reminder:** You may use LLMs (e.g. Grok, ChatGPT, Claude) for coding assistance or proofreading. Any such use must be acknowledged in a footnote or appendix. For example, if using an LLM for code debugging, include a footnote like: 'ChatGPT assisted in optimizing the lm() call on [date]'. Failure to declare will result in a 0 for the affected question.

Hint: Summarising results in tables and including plots makes comparison clearer and easier.

2 Learning Objectives

By completing this assignment you will be able to:

- Explain multicollinearity and overfitting in **OLS** regression and show how regularisation (**Ridge**, **LASSO**, and **ElasticNet**) mitigate these issues.
- Compare subset selection (backwards/stepwise) with penalised regression on sparsity, computational efficiency, and test-set performance.
- Correctly apply **Ridge** ($\alpha = 0$), **LASSO** ($\alpha = 1$), and **ElasticNet** ($0 < \alpha < 1$) using **glmnet**, with proper standardisation and cross-validation tuning of λ (and α).
- Tune both α and λ for **ElasticNet** and justify choosing $lambda.min$ versus $lambda.1se$ using bias–variance trade-offs.
- Implement reproducible workflows (consistent seeds, splits, scaling) and visualise coefficient paths plus CV curves for insight into shrinkage and selection.
- Construct comparison tables and critically evaluate models on test MSE/RMSE, number of predictors, and time across all techniques.
- Interpret selected/shrunk coefficients in domain context (housing prices, Parkinson's speech features, rainfall) and discuss when **Ridge** is preferable to **LASSO** (or vice versa).

3 Question 1

(40 **marks)** In the folder for Assignment 2, question 1 there are 2 files: the RData file **Residen** and the excel file **Residential-Building-Data-Set.xlsx**. **Residen** is a copy of the dataset in the excel file where the variables names have been changed for use in R. The excel file also contains a description of the variables. More information on the dataset can be obtained on the UCI webpage:

https://archive.ics.uci.edu/ml/datasets/Residential+Building+Data+Set.

a). (8 marks) Explore the correlation between the variables using appropriate graphical tools (e.g. **corrplot**). Comment on any strong correlations or groups of correlated predictors.

b). (8 marks) Please fit a linear regression model to explain the "actual sales price" (V104) in terms of the of the other variables excluding the variable "actual construction costs" (V105). Show and interpret the **summary()** output. Randomly split the data using a 80 : 20 split (train/test). Report the baseline test-set MSE.

c). (8 marks) Using the same train/test split used in (b), fit a linear regression model to explain the "actual sales price" (V104) in terms of other variables (**excluding** the variable "actual construction costs" (V105)) using (i) **backwards selection** and (ii) **stepwise selection**. Compare the two models on: outputs, computational time (system.time), number of predictors kept, and test-set MSE.

d). (8 marks) Using the same train/test split used in (c) fit a linear regression model to explain the "actual sales price" (V104) in terms of other variables (**excluding** the variable "actual construction costs" (V105)) using (i) **Ridge regression** (ii) **LASSO regression** with an appropriate λ determined by 10-fold cross validation. Compare these two models in terms of outputs, computational time, test-set MSE, and number of predictors kept. Show the CV error curve for both.

e). (8 marks) Please give a possible explanation of why one method is better than the other in this case, considering also results from backwards and stepwise selection. Refer to the correlation plot, multicollinearity, and bias-variance trade-off.

4 Question 2

(20 **marks)** Also included in the assignment folder is the file **parkinsons.csv**, this dataset [1] contains information on 42 patients with Parkinson's disease. The outcome of interest is **UPDRS**, which is the total unified Parkinson's disease rating scale. The first 96 features, **X1–X96**, have been derived from audio recordings of speech tests (i.e. there has already been a process of **feature extraction**) while feature **X97** is already known to be informative for **UPDRS**.

Read in the dataset and set up the model matrix **X**. Next, randomly split the data into a training set with 30 patients and a test set with 12 patients – remembering to state what your RNG seed was.

Standardize your training and test sets before your analysis: $X = scale(X)$ so that the absolute size of the estimated coefficients will give a measure of the relative importance of the features in the fitted model.

a). (5 marks) Confirm that a linear model can fit the training data exactly. Why is this model not going to be useful for prediction?

b). (8 marks) Now use the **LASSO** to fit the training data, using leave-one-out cross-validation to find the tuning parameter λ. (This means $nfolds = 30$. You will also find it convenient to set $grid = 10\hat{}seq(3, -1, 100)$ and $thresh = 1e - 10$). Report the optimal value of λ and what is the resulting test-set MSE and computational time. Compare to the **lm()** baseline.

c). (7 marks) State your final model for the **UPDRS**. How many features have been selected? Interpret the non-zero coefficients in the Parkinson's disease context. What conclusions can you draw about speech features?

[1] This dataset has been derived from the Oxford Parkinson's Disease Telemonitoring Dataset available at http://archive.ics.uci.edu/ml/datasets/Parkinsons+Telemonitoring

5 Question 3

(40 **marks**) Another dataset included in the assignment folder is the file **Weather_Station_Data_v1.csv**, this dataset contains 2000 observations of weather station data, where the MEAN_ANNUAL_RAINFALL is given against thirteen attributes (features) of the weather stations.

There are no missing or undefined values in the dataset.

The first row of the CSV contains the variable names.

Randomly split the data using a $80:20$ split (train/test).

Train an **ElasticNet** model to predict **MEAN_ANNUAL_RAINFALL** using 10-fold cross-validation to optimize values for α and λ.

Plot the cross-validation results for your model.

Make predictions on your test set using both **lambda.min** and **lambda.1se**, show the coefficients of each model (the non-zero ones), report the MSE/RMSE of both, and the number of predictors used in each model and computational time. Compare to a **lm()** baseline.

Which model (*min* vs 1*se*) would you choose? Justify using bias-variance considerations and the number of predictors retained.

6 The End

Have you answered all the questions?

7 Reproducibility Statement

All analyses were performed in R version … using the following packages: …

```r
sessionInfo()
```

```
R version 4.5.0 (2025-04-11)
Platform: x86_64-pc-linux-gnu
Running under: Ubuntu 20.04.6 LTS

Matrix products: default
BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/liblapack.so.3;  LAPACK version 3.9.0

locale:
```

```
 [1] LC_CTYPE=en_NZ.UTF-8       LC_NUMERIC=C
 [3] LC_TIME=en_NZ.UTF-8        LC_COLLATE=en_NZ.UTF-8
 [5] LC_MONETARY=en_NZ.UTF-8    LC_MESSAGES=en_NZ.UTF-8
 [7] LC_PAPER=en_NZ.UTF-8       LC_NAME=C
 [9] LC_ADDRESS=C               LC_TELEPHONE=C
[11] LC_MEASUREMENT=en_NZ.UTF-8 LC_IDENTIFICATION=C

time zone: Pacific/Auckland
tzcode source: system (glibc)

attached base packages:
[1] stats     graphics  grDevices utils     datasets  methods   base

loaded via a namespace (and not attached):
 [1] compiler_4.5.0    fastmap_1.2.0     cli_3.6.5         tools_4.5.0
 [5] htmltools_0.5.8.1 rstudioapi_0.17.1 yaml_2.3.10       rmarkdown_2.29
 [9] knitr_1.50        jsonlite_2.0.0    xfun_0.53         digest_0.6.37
[13] rlang_1.1.6       evaluate_1.0.5
```

8 Appendix

Don't forget to include your AI use acknowledgement (if applicable).